

Force

In physics, a **force** is a push or pull acting upon an object as a result of the object's interaction with another object.

It is an external agent capable of altering an object's state of rest or motion, causing it to accelerate, decelerate, or change direction.

Key Characteristics

- **Vector quantity** – Forces have both magnitude (strength) and direction.
- **Standard unit** – The metric unit for force is the **Newton (N)**.

One Newton is the force required to accelerate a 1 kg mass at a rate of **1 m s^{-2}** .

- **Formula** – Newton's Second Law of Motion:

[

$$F = m a$$

]

where

* (F) = force,

* (m) = mass,

* (a) = acceleration.

Common Types of Forces

Forces are generally grouped into two main categories:

1. Contact Forces

Applied through physical contact between objects. Examples include:

- **Friction** – resistance to relative motion.
- **Tension** – pulling force transmitted through a string, rope, or cable.
- **Air resistance** – drag force opposing motion through air.
- **Normal force** – perpendicular reaction force from a surface.
- **Push/Pull** – direct mechanical forces applied by hand or machinery.

2. Non-Contact (Field) Forces

Act over a distance without physical contact. Examples include:

- **Gravitational force** – attraction between masses.
- **Electrostatic force** – attraction or repulsion between charged particles.
- **Magnetic force** – interaction between magnetic fields or moving charges.
- **Nuclear forces** – strong and weak forces binding atomic nuclei.

These notes provide a concise, well-formatted overview of the concept of force, its essential properties, the governing equation, and the primary categories of forces encountered in physics.